PassFrame tuition

Perimeter & Area

Year 4 Pt 2 — Mark Scheme — Lesson Worksheet & Homework (tutor copy)

Lesson Worksheet — Answers

Easy / Standard

1. 28cm

2. 36cm

3. 44cm²

4. C — both A (30cm) and B (30cm)

5. 34cm

6. 7cm

7. D — 49cm²

8. 10cm

9. A, B, C, D and E (all of these have a perimeter of exactly 26cm)

10. 34cm

11. 42cm²

12. B — 6×6=36cm² is larger than A's 7×5=35cm²

13. 44cm

14. A square; 64cm²

15. 8cm

Standard

1. 12cm²

2. C — both A (15cm²) and B (15cm²)

3. 30cm²

4. 28 − 4 = 24cm²

5. 96cm

6. B — 24cm²

7. 54m²

8. A and B (areas 20cm² and 24cm² are both more than 18cm²; C=15cm², D=18cm² exactly and E=10cm² are not)

9. 6cm

10. 8400cm²

11. 18cm²

12. 100m²

13. 30cm

14. 70cm² (width 7cm × length 10cm)

15. 60 − 6 = 54m²

Standard / Hard

1. 25×12=300m²; inner = 23×10=230m²; path = 70m²

2. 126 − 12 = 114m²

3. C — a 5cm×9cm rectangle has the largest area (45cm²; the others are 40cm², 42cm² and 36cm²)

4. 6cm (area 36cm² ÷ 12 × 2)

5. £264

6. 48m

7. B and C (area = ½×14×6 = 42cm², which is half the rectangle's 84cm²)

8. 384 − 64 = 320cm²

9. Length 10cm, width 5cm

10. 40m (the same as the original rectangle's perimeter)

11. perimeter of 64m×44m = 216m

12. 20cm (100 ÷ 5)

13. Area = 20+4 = 24m²; cost = £312

14. 80cm² (area is 4 times as large)

15. 64cm² (height 8cm; ½×16×8)

Homework — Answers

Easy / Standard

1. 24cm

6. 6cm

11. 26cm²

2. 32cm

3. 48cm²

4. C — both A (28cm) and B (28cm)

5. 36cm

7. C — 36cm²

8. 9cm

9. A, B, C, D and E (all have a perimeter of exactly 24cm)

10. 30cm

12. B — 25cm² is larger than A's 24cm²

13. 40cm

14. A square; 49cm²

15. 8cm

Standard

1. 20cm²

2. C — both A (24cm²) and B (24cm²)

3. 36cm²

4. 40 − 6 = 34cm²

5. 110cm

6. B — 28cm²

7. 56m²

8. A and D (areas 22.5cm² and 25cm² are more than 20cm²)

9. 6cm

10. 9000cm²

11. 32cm²

12. 121m²

13. 24cm

14. 88cm² (width 8cm × length 11cm)

15. 84 − 12 = 72m²

Standard / Advanced

1. 30×14=420m²; inner = 28×12=336m²; path = 84m²

2. 160 − 15 = 145m²

3. B — a 7cm×8cm rectangle has the largest area (56cm²; the others are 54cm², 54cm² and 49cm²)

4. 10cm (area 50cm² ÷ 10 × 2)

5. £455

6. 52m

7. B and C (area = ½×16×5 = 40cm², half of the 80cm² rectangle)

8. 504 − 36 = 468cm²

9. Length 12cm, width 6cm

10. 44m (the same as the original rectangle's perimeter)

11. perimeter of 74m×49m = 246m

12. 27cm (81 ÷ 3)

13. Area = 24+6 = 30m²; cost = £420

14. 96cm² (area is 4 times as large)

15. 54cm² (height 6cm; ½×18×6)